Math 271 Mathematical Methods in Economics, Fall 2026

Tutorial - 1 *Solution Guideline*

Topics: Preliminaries, Vectors

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1. Let U be the universal set in a given context. Let P be a property for elements in U such that for each $x \in U$ either P holds, in which case we write, $P(x)$; or P doesn't hold in which case we write $\neg P(x)$. Let $P_T \subsetneq U$ be the subset for which P holds. That is, $P_T := \{x \in U : P(x)\}$ Prove the following :

   (a) $[P \implies Q] \iff [P_T \subseteq Q_T]$

   (b) $[P_T \subseteq Q_T] \iff [P_T^C \cup Q_T = U]$ and hence conclude,

$$[P \implies Q] \iff [\sim P \ \vee Q]$$

Answer 1

$\boxed{Part \ \text{(a)} \ [\text{P} \implies Q] \iff [P_T \subseteq Q_T]}$

Consider $(\Rightarrow)$

Suppose $x \in P_T$

$$x \in P_T \iff P(x) \implies Q(x) \iff x \in Q_T$$

Hence, $[x \in P_T \implies x \in Q_T] \iff P_T \subseteq Q_T$

Consider $(\Leftarrow)$

Suppose $P(x)$ holds. Then, $P(x) \iff x \in P_T \subseteq Q_T \iff x \in Q(x)$. Thus,

$$(\forall x \in U) : P(x) \implies Q(x)$$

$\boxed{Part \ \text{(b)} \ [\text{P}_T \subseteq Q_T] \iff [P_T^C \cup Q_T = U]}$

Consider $(\Rightarrow)$

As $x \in U \iff x \in P_T \vee x \in P_T^C$

Case 1: $x \in P_T$

As $P_T \subseteq Q_T$, we have $x \in Q_T$

Case 2: $x \notin P_T$

$x \notin P_T \iff x \in P_T^C$

Combining both cases, $[x \in U] \implies [x \in Q_T \vee x \in P_T^C] \iff [x \in Q_T \cup P_T^C]$

[P.T.O.]

Thus, $U \subseteq Q_T \cup P_T^C$

Also, $Q_T \cup P_T^C \subseteq U$. Hence $Q_T \cup P_T^C = U$

Consider $(\Leftarrow)$

Consider $x \in P_T \iff x \notin P_T^C \iff x \in U \setminus P_T^C \iff x \in Q_T$

Thus, $[x \in P_T \implies x \in Q_T] \implies P_T \subseteq Q_T$

Hence, $[P_T^C \cup Q_T = U] \iff [P_T \subseteq Q_T] \iff [P \implies Q]$

For any $x \in U$, we have either $x \in P_T^C \implies \neg P(x)$, or, $x \in Q_T \implies Q(x)$, hence,

$$(\forall x \in U) : \neg P(x) \vee Q(x)$$

.

Hence, $[P \implies Q] \iff [\neg P \vee Q]$

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2. Suppose $\mathfrak{E}$ and $\mathbb{N}$ denotes the set of all even natural numbers and natural numbers respectively. Prove that if $p \in \mathbb{N}$ and $p^2 \in \mathfrak{E}$, then $p \in \mathfrak{E}$ using
   (a) direct proof
   (b) method of contradiction
   (c) method of contraposition

Answer 2

$\boxed{\textit{Part (a) Direct Proof}}$

Let $p^2 \in \mathfrak{E}$. Then, $p^2|2$. Furthermore, $p^2|p$. Hence, we have,

$$p^2|p \wedge p^2|2 \implies p^2 = 2pk, k \in \mathbb{Z}$$

Thus, $p^2 - 2pk = 0 \iff p(p-2k) = 0 \iff [p = 0 \in \mathfrak{E}] \vee [p = 2k \in \mathfrak{E}]$

In either case $p \in \mathfrak{E}$

$\boxed{\textit{Part (b) Proof by Contradiction}}$

Suppose $p \notin \mathfrak{E}$. Hence, $(p+1) \in \mathfrak{E}$ as well as $(p-1) \in \mathfrak{E}$

As $p^2 \in \mathfrak{E}$, therefore,

$p^2 - (p+1) \in \mathfrak{E}$

Thus, $p^2 - p - 1 = 2k'$ where $k' \in \mathbb{Z}$

$$\implies p^2 - p = 2k + 1 \implies p(p-1) = 2k + 1 \implies p - 1 \notin \mathfrak{E}$$

Hence, $(p-1) \in \mathfrak{E} \wedge (p-1) \notin \mathfrak{E}$, a contradiction.

Part (c) Proof by Contrapositive

Let $p \notin \mathfrak{E}$

RTP: $p^2 \notin \mathfrak{E}$

As $p \notin \mathfrak{E}$, we have $p = 2k+1$ for some $k \in \mathbb{Z}$

$$\begin{aligned} p^2 &= (2k+1)(2k+1) \\ &= 4k^2 + 4k + 1 \\ &= 2(2k^2 + 2k) + 1 \\ &= 2m + 1 \qquad \text{where } m = 2k^2 + 2k \in \mathbb{Z} \end{aligned}$$

Hence, $p^2 \notin \mathfrak{E}$

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3. Consider an initial investment of s in a T-period fixed deposit that pays $s(1+r)^{T-1}$ after T periods. This can be represented as a T-vector, viz., $b = [-s, 0, 0, \ldots, 0, s(1+r)^{T-1}]$. Express b as a linear combination of $(T-1)$ one period unit loans where a one-period unit loan beginning in period $1 \leq t \leq T-1$ is denoted by $l^t = [0, \ldots, 0, -1, (1+r), 0, \ldots, 0]$.That is, find real numbers $\beta_1, \beta_2, ..., \beta_{T-1}$ such that $b = \beta_1 l^1 + \beta_2 l^2 + ... + \beta_{T-1} l^{T-1}$. Convince yourself that the answer corresponds to the idiom "*To borrow from Peter to pay Paul*".

Answer 3

We can write b as follows: $b = sl^1 + s(1+r) + \ldots s(1+r)^{T-1} l^{T-1}$. This can be viewed as borrowing an amount of s in period t from a borrower identified as (s,t) and paying back the borrower (s,t) using a loan from borrower $(s(1+r), t+1)$

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4. Suppose $B = \{b^1, b^2, ..., b^n\}$ form a basis of $\mathfrak{R}^n$. Further, suppose $0 \neq c = \alpha_1 b^1 + \alpha_2 b^2 + ... + \alpha_n b^n$ is a non-zero vector in $\mathfrak{R}^n$. Prove that for any basis vector b^r for which $\alpha_r \neq 0$, is removed from the basis set B and in its place c is added, then $(B \setminus \{b^r\}) \cup \{c\}$ is a set of basis for $\mathfrak{R}^n$

Answer 4

Given $B := \{b^1, b^2, ..., b^n\}$ is a basis for $\mathfrak{R}^n$, hence, we have,

$$\sum_{i=1}^{n} \lambda_i b^i = 0 \qquad \text{with} \qquad \lambda_i = 0 \ \forall i \in \{1, ..., n\}$$

Consider the homogenous system,

$$\sum_{i=1}^{r-1} \mu_i b^i + \mu_r c + \sum_{j=r+1}^{n} \mu_j b^j = 0$$

Suppose, for contradiction, that $\{b^1, b^2, ..., b^{r-1}, c, b^{r+1}, ..., b^n\}$ is a set of linearly dependent vectors.

Substitute $c = \alpha_1 b^1 + \alpha_2 b^2 + ... + \alpha_n b^n$ in the above equation. Thus, we get,

$$\sum_{i=1}^{r-1} \mu_i b^i + \mu_r \left(\sum_{i=1}^{n} \alpha_i b^i \right) + \sum_{j=r+1}^{n} \mu_j b^j = 0$$

[P.T.O.]

$$\iff \sum_{i=1}^{r-1}(\mu_i + \mu_r\alpha_i)b^i + \alpha_r b^r + \sum_{j=r+1}^{n}(\mu_j + \mu_r\alpha_j)b^j = 0$$

$$\iff \sum_{i\neq r}(\mu_i + \mu_r\alpha_i)b^i + \mu_r\alpha_r b^r = 0$$

As B forms a basis for $\mathfrak{R}^n$, we have, $\mu_i + \mu_r\alpha_i = 0$ for all $i \neq r$ and $\mu_r\alpha_r = 0$. Since, $\alpha_r \neq 0$, we have, $\mu_r = 0$. With $\mu_r = 0$, we have $\mu_i = 0$ for all $i \neq r$. Thus, $\forall i \in \{1, ..., n\}$, we have $\mu_i = 0$. Thus, $\{b^1, b^2, ..., b^{r-1}, c, b^{r+1}, ..., b^n\}$ forms a basis for $\mathfrak{R}^n$

Aliter: Consider the basis B. Then, as B is a set of linearly independent vectors, therefore, $B\setminus\{b^r\}$ is a set of linearly independent vectors as well. We claim, $(B\setminus\{b^r\})\cup\{c\}$ is a set of n linearly independent vectors and, hence, a basis for $\mathfrak{R}^n$. As $(B\setminus\{b^r\})$ is a set of linearly independent vectors, therefore, $(B\setminus\{b^r\})\cup\{c\}$ is linearly independent if there exists no non-zero linear combination representation of $c = \gamma_1 b^1 + \gamma_2 b^2 + \cdots + \gamma_{r-1}b^{r-1} + \gamma_{r+1}b^{r+1} + \cdots + \gamma_n b^n$ with $\gamma_i \neq 0$ for all indices $i \in \{1, 2, ..., r-1, r+1, ..., n\}$ (Appendix 1, Theorem 1). We now prove that indeed this is the case, that is there exist no non-zero linear combination representation of the n vector $c = \gamma_1 b^1 + \gamma_2 b^2 + \cdots + \gamma_{r-1}b^{r-1} + \gamma_{r+1}b^{r+1} + \cdots + \gamma_n b^n$ with $\gamma_i \neq 0$ for all indices $i \in \{1, 2, ..., r-1, r+1, ..., n\}$. We proceed by contradiction. Suppose that there is such non-zero linear combination representation

$$c = \gamma_1 b^1 + \gamma_2 b^2 + \cdots + \gamma_{r-1}b^{r-1} + \gamma_{r+1}b^{r+1} + \cdots + \gamma_n b^n$$

$$c = \gamma_1 b^1 + \gamma_2 b^2 + \cdots + \gamma_{r-1}b^{r-1} + \gamma_{r+1}b^{r+1} + \cdots + \gamma_n b^n + 0b^r$$

As per the question, c is given by

$$c = \alpha_1 b^1 + \alpha_2 b^2 + ... + \alpha_n b^n$$

As the representation of any vector $\hat{c}$ is unique (Appendix 1, Theorem 2) with respect to any basis $\hat{B}$, we have that $c = \alpha_1 b^1 + \alpha_2 b^2 + ... + \alpha_n b^n$, whence, $\alpha_r \neq 0$, however, by our supposition, $\alpha_r = 0$, a contradiction. Thus, $(B\setminus\{b^r\})\cup\{c\}$ is a basis for $\mathfrak{R}^n$.

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5. Let $S := \{x^1, x^2, x^3\} \subset \mathfrak{R}^3$ be a set of linearly independent vectors. Let $S' = \{y^1, y^2, y^3\}$ be a new set of linearly independent set of vectors, generated from S as follows:

   (a) $y^1 := \alpha_{11}x^1$

   (b) $y^2 := \alpha_{21}x^1 + \alpha_{22}x^2$

   (c) $y^3 := \alpha_{31}x^1 + \alpha_{32}x^2 + \alpha_{33}x^3$

   Give a sufficient condition on α_{ij} where $i, j \in \{1, 2, 3\}$ for S' to be a set of linearly independent vectors

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Answer 5

The set $S' = \{y^1, y^2, y^3\}$ is a set of linearly independent vectors if and only if we have

$$\beta_1 y^1 + \beta_2 y^2 + \beta_3 y^3 = 0 \implies \beta_1 = \beta_2 = \beta_3 = 0$$

That is, definitionally,

$$[S' \quad \text{is a set of linearly independent vectors}] \iff [\beta_1 y^1 + \beta_2 y^2 + \beta_3 y^3 = 0 \implies \beta_1 = \beta_2 = \beta_3 = 0]$$

For sufficiency, we have to provide a condition, say (C) such that,

$$(C) \implies [\beta_1 y^1 + \beta_2 y^2 + \beta_3 y^3 = 0 \implies \beta_1 = \beta_2 = \beta_3 = 0]$$

I claim (C) is given by $(C) : \alpha_{11} \neq 0, \alpha_{22} \neq 0, \alpha_{33} \neq 0$.

Proof. We now show that the claimed (C) is sufficient. Suppose (C) holds. Substituting for y^1, y^2 and y^3 on the RHS we get,

$$\beta_1 y^1 + \beta_2 y^2 + \beta_3 y^3 = \beta_1(\alpha_{11}x^1) + \beta_2(\alpha_{21}x^1 + \alpha_{22}x^2 + \beta_3(\alpha_{31}x^1 + \alpha_{32}x^2 + \alpha_{33}x^3)$$

Simplifying we get,

$$\beta_1 y^1 + \beta_2 y^2 + \beta_3 y^3 \ = \ (\beta_1\alpha_{11} + \beta_2\alpha_{21} + \beta_3\alpha_{31})x^1 + (\beta_2\alpha_{22} + \beta_3\alpha_{32})x^2 + \beta_3\alpha_{33}x^3$$

Hence, the sufficient condition (C) for S' to be linearly independent is equivalently expressed as

$$(C) \implies [\gamma_1 x^1 + \gamma_2 x^2 + \gamma_3 x^3 = 0 \implies \gamma_1 = \gamma_2 = \gamma_3 = 0]$$

where

(a) $\gamma_1 = \beta_1\alpha_{11} + \beta_2\alpha_{21} + \beta_3\alpha_{31}$

(b) $\gamma_2 = \beta_2\alpha_{22} + \beta_3\alpha_{32}$

(c) $\gamma_3 = \beta_3\alpha_{33}$

As $S = \{x^1, x^2, x^3\}$ is a set of linearly independent vectors, we have

(1) $\gamma_3 = \beta_3\alpha_{33} = 0$. Given our sufficient condition, $(C) : \alpha_{11} \neq 0, \alpha_{22} \neq 0, \alpha_{33} \neq 0$, we get $\beta_3 = 0$

(2) whence, $\beta_2\alpha_{22} = 0$. From this, and (C), we get $\beta_2 = 0$

(3) whence, $\beta_1\alpha_{11} = 0$. From this, and (C), we get $\beta_1 = 0$

Intuitively, for $\{y^1, y^2, y^3\}$ to be a set of linearly independent vectors, it couldn't be the case that $\alpha_{11} = 0$, since, $\alpha_{11} = 0 \implies y^1 = 0$. Similarly, we it couldn't be the case that $\alpha_{22}\bar{0}$, else, $y^2 = ky^1$ where $k = \frac{\alpha_{21}}{\alpha_{11}}$. Similarly, it couldn't be the case $\alpha_{33} \neq 0$, else, $y^3 = k'y^1 + k''y^2$ where $\frac{\alpha_{31}\alpha_{22} - \alpha_{32}\alpha_{21}}{\alpha_{22}\alpha_{11}} = k'$ and $k'' = \frac{\alpha_{32}}{\alpha_{22}}$. Thus collecting these conditions together in $(C) : \alpha_{11} \neq 0$, $\alpha_{22} \neq 0$,$\alpha_{33} \neq 0$, and substituting (C) in $(c), (b)$ and (a), respectively, we get $\beta_1 = \beta_2 = \beta_3 = 0$. Hence, $\alpha_{11} \neq 0, \alpha_{22} \neq 0 \quad$ and $\quad \alpha_{33} \neq 0$ is indeed a sufficient condition for linear independence for S'.

□

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6. Let $x^1, x^2 \in \Re^d$ be non-zero vectors ($x^1, x^2 \neq 0$). Define the harmonic mean of their squared norms as:

$$H(x^1, x^2) := \frac{2\|x^1\|^2 \|x^2\|^2}{\|x^1\|^2 + \|x^2\|^2}$$

Suppose x^1 and x^2 satisfy the dot-product condition:

$$0 \leq x^1 \cdot x^2 < H(x^1, x^2)$$

(a) **(AM–HM Bound)** Show that $H(x^1, x^2) \leq \frac{\|x^1\|^2 + \|x^2\|^2}{2}$, with equality if and only if $\|x^1\| = \|x^2\|$.

(b) **(Linear Independence)** Prove that $\{x^1, x^2\}$ is a linearly independent set.

Answer 6

$\boxed{\textit{Part (a)} \quad \textbf{(AM-HM Bound)}}$

$$
\begin{aligned}
& RTP: & H(x^1, x^2) &\leq \tfrac{\|x^1\|^2+\|x^2\|^2}{2} \\
\iff & RTP: & \tfrac{\|x^1\|^2+\|x^2\|^2}{2} - \tfrac{2\|x^1\|^2\|x^2\|^2}{\|x^1\|^2+\|x^2\|^2} + &\geq 0 \\
\iff & RTP: & \tfrac{\left(\|x^1\|^2+\|x^2\|^2\right)^2-4\|x^1\|^2\|x^2\|^2}{2\|x^1\|^2+\|x^2\|^2} &\geq 0 \\
\iff & RTP: & \tfrac{\left(\|x^1\|^2-\|x^2\|^2\right)^2}{\|x^1\|^2+\|x^2\|^2} &\geq 0
\end{aligned}
$$

which is always true, and it holds with equality if $\|x^1\|^2 = \|x^2\|^2 \iff \|x^1\| = \|x^2\|$

$\boxed{\text{Part } (b) \quad \textbf{(Linear Independence)}}$

Consider the equation,

$$\lambda_1 x^1 + \lambda_2 x^2 = 0$$

Suppose $\{x^1, x^2\}$ is not a set of linearly independent vectors. Then, there exists $\lambda_i \neq 0$ such that $\lambda_1 x^1 + \lambda_2 x^2 = 0$.

Without loss of generality, let $\lambda_1 \neq 0$, whence, $x^1 = kx^2$ where $k = -\frac{\lambda_2}{\lambda_1}$ Then,

$$x^1 \cdot x^2 = kx^2 \cdot x^2 = k\|x^2\|^2 \geq 0 \iff k \geq 0$$

.

We claim, $\lambda_2 \neq 0$.

Proof. Suppose not. Suppose $\lambda_2 = 0$. Then, $\lambda_1 x^1 + \lambda_2 x^2 = 0 \implies \lambda_1 x^1 = 0$. As $x^1 \neq 0$ and $\lambda_1 \neq 0$, we have $\lambda_1 x^1 \neq 0$, a contradiction. Thus, $\lambda_2 \neq 0$. □

Thus, $x^2 = \frac{1}{k} x^1$ with $k \neq 0$.
Substituting $x^1 = kx^2$ into $x^1 \cdot x^2 < H(x^1, x^2)$:

$$x^1 \cdot x^2 = k\|x^2\|^2 < H(kx^2, x^2) = \frac{2k^2}{k^2+1}\|x^2\|^2 \iff \boxed{k < \frac{2k^2}{k^2+1}}$$

As $k \neq 0$, we have,

$$\frac{1}{k} < \frac{2}{k^2+1} \iff k^2 + 1 - 2k < 0 \iff (k-1)^2 < 0$$

which is impossible.

Appendix

Theorem 1. *Suppose $S = \{x^1, ..., x^n\}$ is a set of n distinct linearly independent vectors then for each index $i \in \{1, ..., n\}$, there is no family scalars $(\alpha_j)_{j \neq i}$ such that $x^i = \sum_{j \neq i} \alpha_j x^j$*

Proof. Fix i and suppose, for contradiction, that such scalars $(\alpha_j)_{j \neq i}$ exist. That is, suppose there exists an index i associated with vector x^i such that there exists scalars $\{\alpha_1, \alpha_2, ..., \alpha_{i-1}, \alpha_{i+1}, ..., \alpha_n\}$ such that

$$x^i = \alpha_1 x^1 + \alpha_2 x^2 + ... + \alpha_{i-1} x^{i-1} + \alpha_{i+1} x^{i+1} + ... + \alpha_n x^n$$

Subtracting RHS from both sides, we get,

$$x^i - \sum_{j \neq i} \alpha_j x^j = 0$$

Define $\beta_i = 1$ and $\beta_j = -\alpha_j$ for $j \neq i$. Then the above equation becomes,

$$\beta_1 x^1 + \beta_2 x^2 + ... + \beta_n x^n = 0$$

Since, S is linearly independent, this forces $\beta_k = 0$ for every $k \in \{1, ..., n\}$. In particular for i, we must have $\beta_i = 0$ but above we have $\beta_i = 1 \neq 0$, a contradiction. Hence no such scalars exist. □

Theorem 2. *For any basis $B = \{b^1, b^2, ..., b^n\}$ of $\mathfrak{R}^n$, there is a unique representation of a vector $c \in \mathfrak{R}^n$ in terms of linear combinations in B.*

Proof. Suppose not. Suppose there is a basis B and a vector $c \in \mathfrak{R}^n$ such that there exists two distinct family of scalars, $(\alpha_i)_{i \in \{1,...,n\}}$ and $(\gamma_i)_{i \in \{1,...,n\}}$ such that

$$c = \alpha_1 x^1 + \alpha_2 x^2 + ... + \alpha_n x^n$$

$$c = \gamma_1 x^1 + \gamma_2 x^2 + ... + \gamma_n x^n$$

Subtracting the two from each other, we get,

$$0 = (\alpha_1 - \gamma_1) x^1 + (\alpha_2 - \gamma_2) x^2 + ... + (\alpha_n - \gamma_n) x^n$$

Define $\beta_i = \alpha_i - \gamma_i$ for all $i \in \{1, ..., n\}$. Then the above equation becomes,

$$0 = \beta_1 x^1 + \beta_2 x^2 + ... + \beta_n x^n$$

As $(\alpha_i)_{i \in \{1,...,n\}} \neq (\gamma_i)_{i \in \{1,...,n\}}$ there exists $j \in \{1, ..., n\}$ such that $\alpha_j \neq \gamma_j \implies \beta_j \neq 0$, a contradiction to B being a basis for $\mathfrak{R}^n$. □